

CoChem-LUMOS: Photochemical Cleavage Analysis

Automated Pipeline Report

August 6, 2026

1 Methodology

Open-shell photo-cleavage was simulated using a 266.0 nm excitation pulse. Wigner phase-space sampling generated the initial conditions, which were subsequently propagated via the AIMNet2 neural network potential.

2 Spin-State Validation

Broken-symmetry DFT was utilized to extract the final EPR tensors. The expectation value of the spin squared operator was explicitly calculated to monitor spin contamination:

$$\langle S^2 \rangle = 0.7501 \tag{1}$$

3 Conclusion

The tensor data has been serialized to `lumos_tensors.npz` and is ready for experimental matching.